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**Project- Microplastic and its
Effect on Water and Aquatic
Environment**

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1. Introduction

Increased productivity and the slow biological breakdown of plastic have detrimental effects on aquatic life. Plastics that enter the marine environment can persist for hundreds of years, fragmenting due to mechanical and photochemical processes and transforming into microplastics (5 mm) or nanoplastics (< 1 m). Plastics are organic polymers generated from petroleum, such as polyethylene, polypropylene, polyvinyl chloride, and polyester. The most prevalent are PE and PP, which account for more than 80% of global output (Leng et al., 2018). PET is predominantly used in packaging due to its safety, lightweight characteristics, affordability, and low manufacturing cost. At 1.37–1.45 g cm³, PET sinks rapidly, making it appropriate for benthic creatures (Weber et al., 2018). While PET is resistant to weathering, fragmentation mechanisms are not, and in marine environments, photooxidation and hydrolysis are likely. The ocean's pH fluctuation may affect the chemical balance of microplastics by increasing or lowering the rate of chemical leaching from their surface (Piccardo et al., 2020). Polymers' ability to adjust their properties has resulted in their widespread use in various domestic and industrial applications (P. M. Gore & Kandasubramanian, 2018b; Inamdar et al., 2018; Jayalakshmi et al., 2020; Kumar et al., 2020). (Gupta et al., 2016; Gupta & Kandasubramanian, 2017; Kavitha & Kandasubramanian, 2020; Sahetya et al., 2015; Sharma et al., 2016), the average particle size of plastic in the atmosphere appears to be decreasing, which has resulted in an increase in microplastic flotsam and jetsam in recent years. Plastics removed from the aquatic environment eliminate a severe condition that has a detrimental effect on the tourism, shipping, trawling, and aquaculture. Microplastics are a prevalent contaminant in the aquatic environment, acting as transporters for pollutants (Rodrigues et al., 2019). Numerous aquatic animals consume microplastics, creating physiological disturbances that harm human health (Fig. 1).

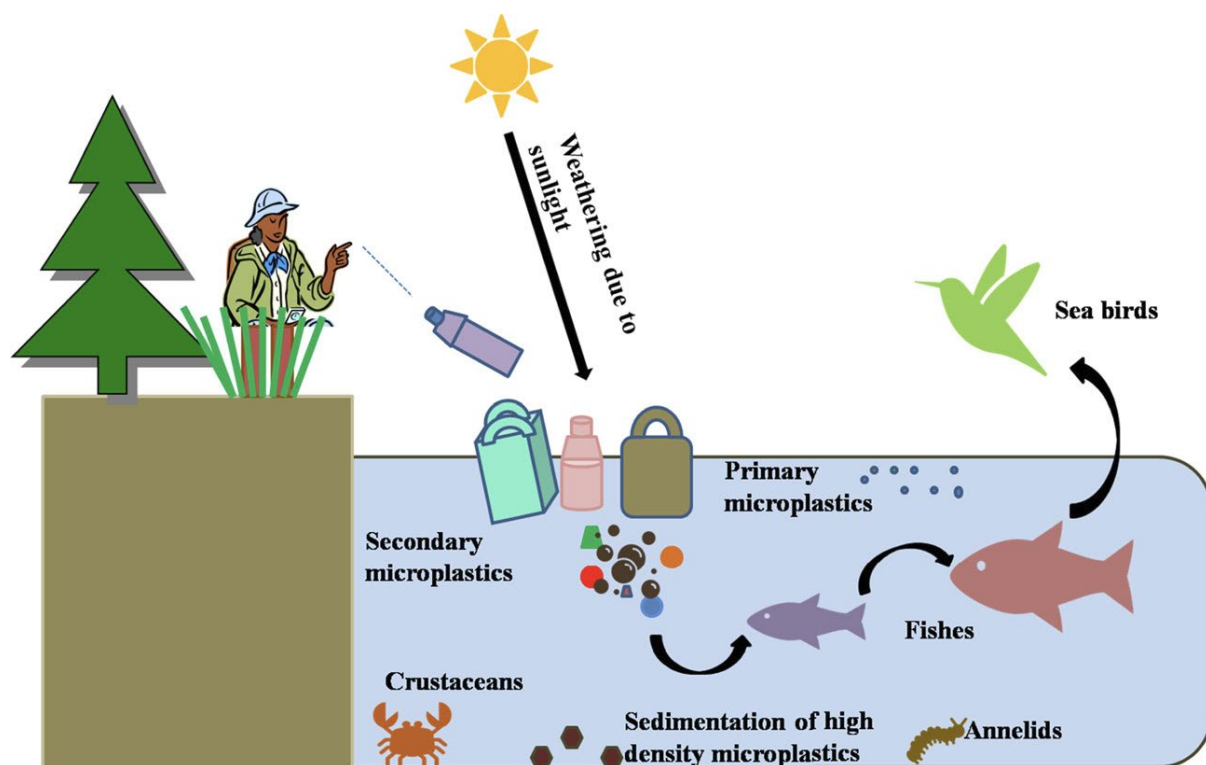


Figure 1: Microplastic pathway in organisms

They are rapidly ingested and excreted by a wide variety of marine creatures, precluding conclusive evidence of biomagnification (Cozar et al., 2014). MP effects include decreased food intake, developmental problems, and behavioral changes. Nearly 700 aquatic species have been damaged by microplastics worldwide, including sea turtles, penguins, and crab (Marn et al., 2020). However, the microplastic problem is diminishing as most sufferers stay undiagnosed. Plastics enter the ecosystem as a result of human error or as unregulated waste from water treatment plants, sewage treatment plants, and textile manufacturers (Ayalew et al., 2012; Gupta & Kandasubramanian, 2017; Rastogi & Kandasubramanian, 2020; Singh & Balasubramanian, 2014). Inadequate landfill burial practices result in the accumulation of terrestrial plastic in water systems (Anbumani & Kakkar, 2018). Plastic pollution is exacerbated by the mass production and dispersion of plastics into the marine ecosystem (Thushari & Senevirathna, 2020). Microplastics of varying sizes create a habitat for microorganisms (Yang et al., 2020). As a result, microplastics can easily gather and release hazardous organic pollutants such as DDT and other manufacturing additives, hence increasing their concentration (Campanale et al., 2020; Gonte & Balasubramanian, 2012; P. Gore et al., 2018; P. M. Gore & Kandasubramanian, 2018a, 2018b; Rajhans et al., 2019; Thakur & Kandasubramanian, 2019). The smaller the particle size, the greater the risk of microplastics causing injury; however, the effects on marine organisms are unknown (Law & Thompson, 2014). Additive-free microplastics do not affect aquatic creatures chemically, they create physical problems such as intestinal obstructions (Udayakumar et al., 2021). Various chemicals are added to virgin microplastics as needed to enable them to adsorb pollutants from water and hence function as vectors. Plastic poses health concerns at every stage of its lifecycle, from fossil fuel extraction to consumer use to disposal, according to studies (P. M. Gore & Kandasubramanian, 2018b; Issac & Kandasubramanian, 2020).

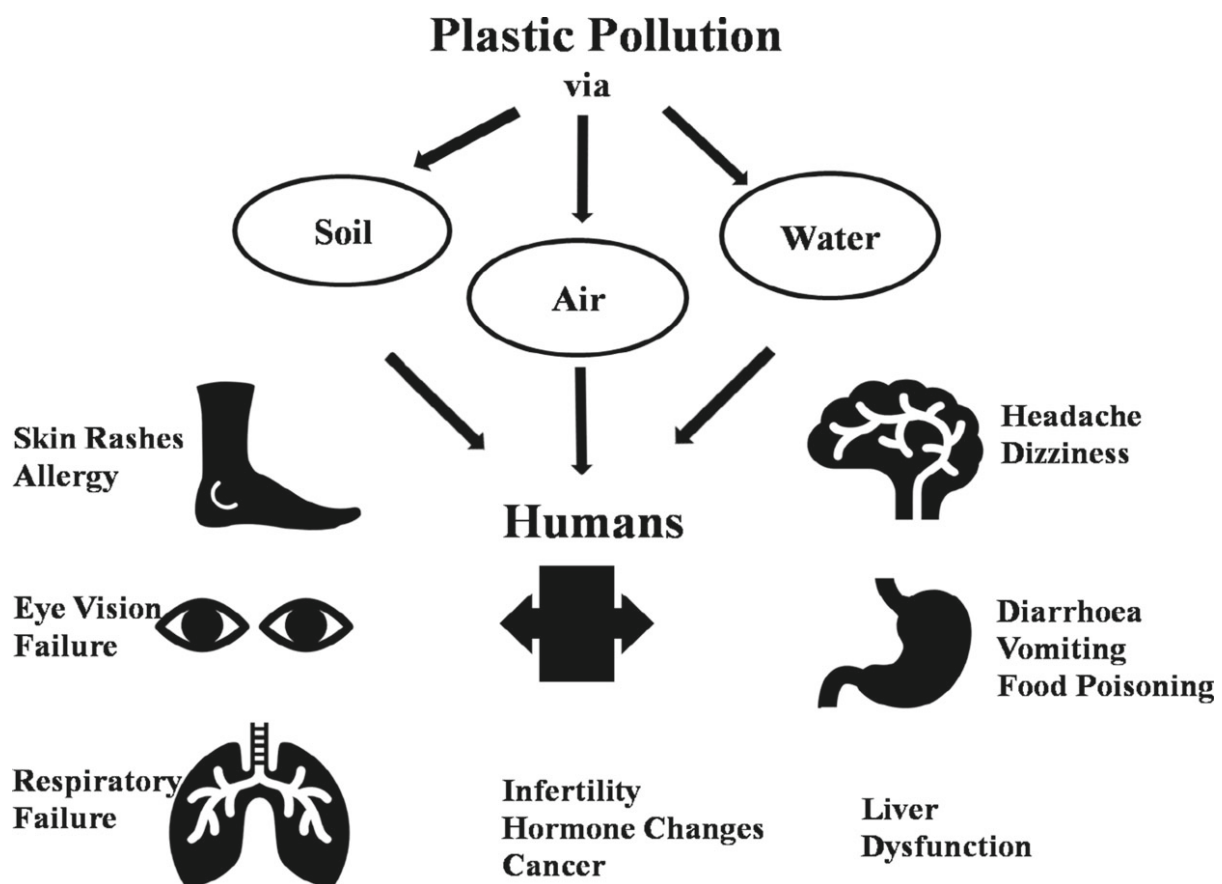


Figure 2: Human Health Implications In Plastic Pollution.

Figure 2 summarises recent human health implications. Because microplastics can harm a wide variety of creatures, the risk to people is not negligible. Humans are the ultimate consumers of microplastic-affected seafood (Saha et al., 2021), so there is a significant danger of microplastic contamination in people (Smith et al., 2018). Microplastics have been detected in drinking water, seawater, bottled water (Selvam et al., 2020). These findings highlight the several routes by which microplastics can infiltrate the human body. Microplastics have been discovered in human faeces and placenta (H. Zhang, 2021). (Ragusa et al., 2021) If plastics can cause such harm to humans, what happens when we swallow such minute, much more deadly particles remains unknown.

This article highlights one of the world's most pressing issues today, namely microplastics. Plastic waste of various sizes, including macro, meso, micro, and nano, is moved across the oceans by waves and winds and is found floating on the surface. When floating microplastics are mistaken for food, they are consumed, wreaking havoc on the health of aquatic species. The most serious problems associated with plastic pollution are determined by the composition of the waste and the pathways it takes to reach the marine ecosystem. Waste discharged from treatment plants, sewer overflows during heavy rains, and biosolid runoff from agricultural areas were all tactics used to contaminate aquatic systems with microplastics. Due to the difficulties associated with identifying and classifying microplastics, research on them is limited. This review discusses the source, distribution, and composition of microplastics, the effects of microplastics in marine and fresh water systems, the role of COVID-19 in microplastic pollution, and the extent to which microplastics affect aquatic organisms, with the goal of raising awareness about the negative effects of microplastic.

2. Microplastics

People have been concerned about the potentially hazardous and obnoxious contamination generated by plastics for ages. Nonetheless, considerable research has been undertaken in recent decades on less apparent plastic waste with less than 5 mm diameter, referred to as microplastics (Weis, 2020). Microplastics are solid particles or polymer matrixes of synthetic origin that are either regular or irregular in shape size from μm to 5 mm. They are insoluble in water (Frias & Nash, 2019). Microplastics come in a variety of forms, including fiber, film, foam, spherical, and pellet (Cowger et al., 2020). (Thompson, 2015) defined microplastics as the accumulation of microparticles with a diameter of less than 20 μm in aquatic systems. Microplastics originate from two distinct sources: one is primary, in the form of nurdles or powders, and the other is secondary, as a result of the fragmentation of bigger particles (Thompson, 2015). Microplastics' shrinking size makes them bioavailable throughout the food chain. Particles of varied sizes undoubtedly have varying impacts, i.e., tiny particles have intrinsically different consequences than large particles, as the particles accumulate within the tissue and disrupt physical processes (Campanale et al., 2020). Microplastics' floatable and persistent qualities enable them to be widely spread in ocean current as a marine contaminant for pollutant transfer to aquatic animals. Microplastics are abundant in aquatic habitats from the poles to the equator at all depths of the sea (Thompson, 2015). Numerous studies have examined the effect of MPs on a variety of marine animals, including mussels (Paul-Pont et al., 2016)

Microplastics' harmful effects are also transmitted to high trophic levels indirectly via the eating of microplastic-injected organisms (Avio et al., 2017). Microplastics' smaller size results in

their infiltration and direct release into water resources during wastewater treatment. In general, microplastics are considered resistant to biotic breakdown. Certain materials are degraded biotically by fungi and bacteria and are ingested or passively absorbed by users at successive trophic levels, leading to gastrointestinal obstruction (Glaser, 2019). Microplastics are found in all stages of the marine food chain (Setälä et al., 2014, 2018). The total amount of MPs consumed varies significantly between organisms and locations, and even within the same region. Aquatic species are well-known for swallowing microplastics in addition to their food, demonstrating that various animals consume microplastics due to their size similarity to their food (Desforges et al., 2015; Walkinshaw et al., 2020). The study's findings indicate that nearly all aquatic organisms consume microplastics. However, the amount consumed varies significantly between species. Predictably, there are three types of adverse consequences associated with microplastic absorption: (1) physiological consequences of intake (Davidson & Dudas, 2016; Pedersen et al., 2020). The larger the number of MPs taken, the greater the danger of adverse effects on the consuming species, such as stunted development and variation in feeding patterns (Horton et al., 2018). Leaching of hazardous compounds into body tissues—Additives such as plasticizers, antioxidants, flame retardants, and pigments used in the manufacture of plastic may leach into body tissues, causing induced alterations or bioaccumulation. Additionally, toxicity can vary depending on the additive ratio required for each material (Botterell et al., 2019). Toxic reaction to pollutants absorbed involuntarily by microplastics—large surface area due to weathering, longer exposure times, and hydrophobic nature all contribute to the sorption of pollutants to the microplastic surface at a higher concentration, thereby acting as a carrier for contaminants to enter aquatic species. Polycyclic aromatic hydrocarbons (PAHs), PCBs, DDT, organohalogenated insecticides, hexachlorocyclohexanes, and chlorinated benzenes are only a few of the pollutants commonly found on microplastics. POPs such as PBDE, PCB, and a few other compounds have been discovered to mimic natural hormones, resulting in reproductive dysfunction. The dynamics of persistent organic pollutants absorption into plastic materials are, of course, dependent on the properties of both the polymer and the contaminant. Microplastics enter the human body through cosmetics, eating habits, dust particles, and the use of plastic objects. The quantity of microplastic in the marine ecosystem continues to grow as a result of continuous increases in plastic manufacture, resulting in harmful repercussions.

3. Numerous microplastics

Microplastic particles are classified according to their dimension, color, content, and density (shown in Fig. 3). Plastic particles of a diameter higher than 25 mm, between 5 and 25 mm, 1 to 5 mm, and 1 nm to 1 µm, respectively, were classified as macro, meso, micro, and nano (Gigault et al., 2018; Lee et al., 2020). Microplastics are classified as primary or secondary based on their origin and intended use. Primary microplastics are produced on a microscopic scale and are found in personal care items such as toothpaste, scrubbers, and other cosmetics (Auta et al., 2017; Duis & Coors, 2016). This type of microplastic skincare product (Napper & Thompson, 2016) supplants a number of commonly used natural cosmetics containing oatmeal, walnuts, or almonds. Small plastic particles, typically 0.25 mm in diameter, are widely employed in cosmetics and industrial abrasive shot-blasting agents. Microplastic particles, such as granules and powders, are employed expressly in a wide variety of applications (Ng et al., 2018; Sharma & Chatterjee, 2017). The MPs demonstrate a range of sizes, i.e., grains of diverse sizes within the same product. Gregory investigated the size variations of microplastics and reported in 1996 that in a similar cosmetic, he discovered PE and PP granules less than 5 mm in diameter and PS spheres less than 2 mm in diameter (Gregory, 1996). Primary microplastics are discharged directly into the environment by houses, factories, and sewerage systems.

Microplastic beads from skincare products are conveyed through the sewage system with wastewater where they are ineffectively removed and consequently accumulate in the ecosystem. Synthetic clothes contain microplastics as fibers release an average of about 700,000 fibers from 6 kg of clothes in a single wash (Napper & Thompson, 2016). Pellets used as a feedstock for plastic goods in industrial applications are also a source of microplastic entering the environment. Microplastics used in dentistry and pharmaceutical carriers enter the environment via wastewaters in the medical profession. Primary microplastics are difficult to remove from aqueous systems due to their small size and limited perceptibility the unswerving discharge of primary microplastics, the debris of larger plastics under the influence of UV and heat can slowly become fragile and then fragment into smaller particles with the help of mechanical forces like winds and ocean currents (Thompson, 2015).

The bulk of microplastics found in aquatic habitats is a result of bigger plastics being broken down, resulting in secondary microplastics (Waller et al., 2017). Fragmentation of bigger polymers is temperature and UV light-dependent. Apart from the fragmentation in the environment, numerous materials split during use, releasing microparticles into the atmosphere, such as the fibers released from clothes during washing. These second microplastics form a gradual deterioration or disintegration of big plastics in the atmosphere caused by ultraviolet light, wave abrasion, or microbial degradation. Microplastics in the environment decay to form nano plastics less than 100 nm in diameter, which have essentially unknown hazards and hazardous qualities compared to other plastic trash (Koelmans, 2015). Due to intense weathering and other mechanical activities, plastics splinter, resulting in more secondary microplastics in aquatic systems than primary microplastics. Microplastics are classified according to their morphology into filaments, microbeads, nurdles, foams, and pieces. Biopolymer synthetic microplastics are less concerning due to their lower hydrophobicity and biodegradability, resulting in carbon dioxide and water formation.

4. Provenance and distribution of microplastics

Microplastics enter aquatic systems from various sources and by a variety of channels (Browne, 2015; Browne et al., 2008). The sources include the manufacturing of plastic products, water, industrial or agricultural wastewater, weathering of plastics, fisheries or aquaculture (Xue et al., 2020; W. Zhang et al., 2017).

Location	Samples collected	MP abundance	MP type	Source	Reference
Dongting Lake (China)	15 each (surface water and sediment)	Surface water: 0.62–4.31 items/m ² Sediment : 21–52 items/100 g dw	50–91% fibres, 5.67–33.33% beads, 2.63–20.00% fragments	Man-made and surrounding rivers	Hu et al. 2020
Kuwait coastal areas	44 (beach sediments)	37 (number)	PP PE PS	Storm sewer and Seaport	Saeed et al. 2020
Northern Tyrrhenian sea (Italy)	58 sediments	1.70 ± 0.93 particles kg ⁻¹	73.8% filaments, 17.2% fragments, 9.0% films	Marine and domestic based	Mistri et al. 2020
Salina Island (Italy)	60 sediments and 5 species	Sediments: 49.0 ± 1.4 items/kg (landslides), 153.5 ± 41.7 items/kg (cliff), 106.0 ± 104.7 items/kg (banks) Holothurian species: 1.3 ± 0.9 items/animal (landslides), 3.8 ± 0.7 items/animal (cliff), 2.47 ± 0.3 items/animal (banks)	Sediments : PVC, PET, PE, PS, PA, PP Holothurian species: PET, PA	Landslides, cliffs and banks	Renzi et al. 2020
Ganges river (India)	120 water samples	71.6% (pre-monsoon), 61.6% (post-monsoon)	91% fiber, 9% fragments	Domestic	Napper et al. 2021

Figure 3: The number, kind, and source of microplastics in different location

The number, kind, and source of microplastics in diverse regions are depicted in Figure 3. Plastic garbage from households, factories, and other sources enters the marine system directly or indirectly through other bodies of water, increasing its volume and damaging aquatic life. Microcapsule fertilizers, which are favored in agriculture to avoid nitrate leaching into groundwater, are a primary source of microplastics that flow out to the oceans via paddy field channels, indicating a greater volume of MP flow during irrigation than during the non-irrigation season. Scratches and browning on the top surface of microcapsules signal the emission of secondary microplastics during the paddy runoff process (Katsumi et al. 2020). According to research done by the United Kingdom's government in 2020, microplastics shed from vehicle tyres of microplastic contamination in the oceans. The tyre is a composite material composed of elastomer, carbon black, fiber, and other organic and inorganic components that contribute to its resilience (Evangelidou et al., 2020). Most tyre particles reach the sea directly via air or other waterwater. Fishing, fish hatcheries, and offshore drilling are all direct sources of plastic that infiltrate aquatic systems and constitute a threat to biota as secondary microplastics following long-term deterioration ((Barboza et al., 2018). Inadequate garbage handling contributed to the Fig 3 Microplastic contamination occurs in freshwater habitats (Free et al., 2014). Smaller size and low density, microplastics are distributed by winds and waves and are hence ubiquitous (Shahul Hamid et al., 2018). Plastic debris carried by wastewater is not effectively removed by treatment plants and hence accumulates in the atmosphere(Akarsu et al., 2020; Naji et al., 2021). Microplastic ingestion can also occur indirectly, with higher creatures in the food chain directly feeding the organisms that mistakenly consume microplastics. Apart from physiochemical mechanisms, the source's nature and entry point also influence the microplastics' distribution in water resources. Source detection is critical for accurately estimating the amount of microplastics entering the marine surface and implementing effective handling procedures. Their distribution throughout the aquatic system can determine Microplastics' potential impacts. Pollution has a lower impact in areas with fewer microplastics and a greater impact in areas where microplastics accumulate. Thus, it is critical to understand how microplastics are distributed to mitigate the looming dangers. Microplastics' distribution and transit are intricately influenced by various elements, including weathering and fragmentation, biofouling, tides, and strong currents. Microplastics are distributed amongst the ocean floor, the water column, the seabed, the coast, and ecology, with each compartment exhibiting distinct biological, physical, and chemical pathways (Choy et al., 2019; Katija et al., 2017). Due to a lack of compartment knowledge, the ramifications and potential for decrease are unknown.

5. Plastics' role in the fate of microplastics.

Microplastics are made up of a complex array of polymers with repeated monomers forming the polymer's backbone. The basic difference between polymers is this backbone structure, which defines a plastic's physicochemical properties (Rochman et al., 2019). Thermoplastic and thermosetting are classified into microplastics. Thermoplastics are a type of plastic that melts when heated and solidifies when cooled. PE, PP, PS, PET, ABS, PC, and PA are all thermoplastic materials. Thermosets are materials that, once cured, stay permanently solid. Examples include polyurethane, urea-formaldehyde, and vinyl ester. Thus, MP does not consist of a single polymer but rather is derived from a wide set of chemically distinct molecules (Rochman et al., 2019). Bioavailability is the fraction of particles in the environment that are available for absorption by an organism (Vallero, 2016). The bioavailability of microplastics is regulated by a range of factors, the most significant of which are the amount and characteristics of plastic. As plastic particles continue to degrade and disintegrate, the amount

of microplastic that becomes physiologically available to species will increase with time (Botterell et al., 2019). On the other hand, microplastics are prone to frequently change density and buoyancy due to biofouling and ingestion, making them accessible to species at varying levels throughout the water column. Benthic species are biologically accessible to higher density plastics, whereas pelagic species are primarily bioavailable to lower density plastics. Thus, the composition of microplastics is a critical aspect. The composition of plastics refers to the type of polymer used, which in turn defines the debris's density. Low-density plastics, such as PP and PE, produce less dense trash than water and are thus likely to float. In contrast, PET, PS, and cellulose acetate contain denser plastics than water and thus appear to settle (Driedger et al., 2015) in terms of their rise and sink velocity. Positively buoyant plastics, which frequently float on the water's surface, will remain there only until they are repeatedly fouled and end up in the benthic zone.

Reduced fouling on sinking plastics caused by grazing can temporarily refloat them, resulting in cyclic floating and sinking until they eventually settle in the ocean depths (Alimi et al., 2021). Positive buoyant plastics are found within the first 100 to 150 meters of the ocean's surface, whereas neutral buoyant plastics are located above 3500 meters from the bottom (Mountford & Morales Maqueda, 2019). All high-density polymers, bio-polluted materials, polymers with fillers, and composites settle. Biofouling caused by the aggregation of microorganisms or algae increases the density of microplastics and contributes to their settling in pelagic or benthic environments. Plastics when deposited in ocean bottom are sheltered from UV light, significantly slowing the breakdown process (Corcoran, 2015). Different grades of polyethylene have varying densities, strengths, crystallinity, and weatherability, with each phase having a specific application, such as LLDPE and LDPE for plastic bags and HDPE for jugs. However, their existence, fate, degradation, and proclivity to sorb/release persistent organic pollutants are determined by the polymers' properties (Andrady, 2017). Semicrystalline polymers (Guo & Wang, 2019) have a higher strength and resistance to fatigue, whereas amorphous polymers have low strength and resistance to fatigue (Sysel, 2016). Plastics such as PE, PP, and PET have a semi-crystalline structure that makes them strong, but at high degrees of crystallinity, the material becomes brittle, contributing to the material's susceptibility to breaking and fracturing during weathering. (Andrady, 2017) MP's characteristic properties, like as crystallinity and density, are not inherent characteristics and can be easily modified with weathering or ageing (Guo & Wang, 2019). Discoloration occurs in PE, PP, PS, PET, PC, and PVC as a result of oxidation, typically due to the accumulation of breakdown products or stabilisers used during production (Gautam et al., 2020; Sharma & Balasubramanian, 2015). Photodegradation in PVC results in the loss of HCl, resulting in the formation of yellow-colored conjugate unsaturation. Due to the low oxygen content and temperature in oceans, MP photodegradation is greatly slowed. Indeed, MP at the ocean surface is more effective against surface fouling than MP on land, protecting them from ultraviolet radiation and resulting in a slower deterioration rate (Guo & Wang, 2019). Figure 5 illustrates how degradation alters the distinctive features of microplastics. The adverse effects of microplastics on microplastic-exposed species can be classified into two categories: physical and chemical products.

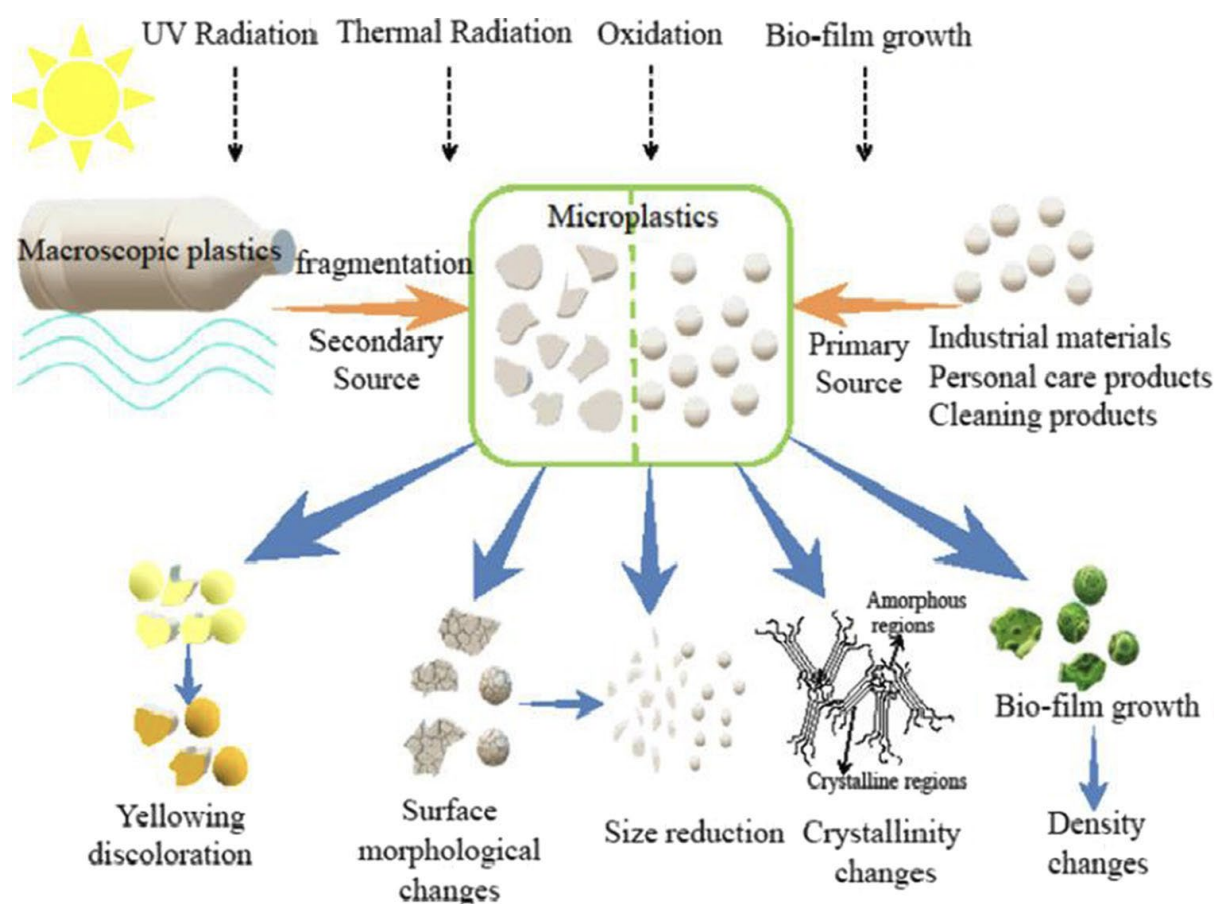


Figure 4: Microplastics Degradation

Physical refers to microplastics' shape, size, and concentration, whereas chemical refers to the hazardous compounds linked with them (Campanale et al., 2020). The rate of sorption of hydrophobic pollutants by microplastics varies according to their shape and polymer type (Tourinho et al. 2019). Microplastics are frequently associated with or in a complex collection of substances, indicating that they can gather organic compounds and trace metals from their environment. MP is primarily composed of additives or raw ingredients derived from plastics and chemicals found in the surrounding environment (Campanale et al., 2020). Due to their high surface area to charge ratio, MPs absorb persistent organic pollutants and other inorganic contaminants. When marine creatures consume MPs, the toxicity increases due to the aggregation of organic hydrophobic molecules. Polymer's physical and chemical properties, such as diffusivity, surface area, crystalline structure, and hydrophobic nature, dictate the amount and type of chemicals that the plastic will accumulate ((Rochman et al., 2019). Rubbery polymers such as polyethylene and polypropylene are supposed to diffuse more chemicals than PET and PVC glass polymers. PE is frequently more receptive to contaminants than other polymers (Robeson et al., 2015) . Except PS glass polymer (Hüffer & Hofmann, 2016) exhibit sorption rate due to the presence of a benzene ring that facilitates chemical addition to the polymer, demonstrating the relationship between sorption and the chemistry of plastics (Alimi et al., 2021). When aquatic species consume MP, it bioaccumulates, causing disturbances in their metabolism, feeding behaviours, development, and reproduction, with varying degrees of toxicological risk for each species. For instance, polystyrene decreased chlorophyll concentrations in algae (Hazeem et al., 2020), whereas *Daphnia magna* exhibited reproductive cycle disruption.

6. Microplastics' effects on numerous organisms

The accumulation of contaminants at trophic levels results in the effective transmission of pathogens across the food chain. Retention of plastic debris within organisms may result in chemical leakages if any additives are present, resulting in cumulative effects (Setälä et al., 2014). Microplastics prevalent in marine systems worldwide have an effect on the feeding, growth, spawning, and survival of aquatic organisms. However, the extent to which microplastics disrupt higher complex food chains via transmitting chemicals contained in and on the surface of MP is unknown (Granek et al., 2020). Due to the scarcity of data on trophic transfer (Furtado et al., 2016), it is unknown if contaminants are expelled or bioaccumulated at higher trophic levels. The cumulative consequence of microplastic injections is a decrease in aquatic species' eating ability; other issues include effects on growth and Fig. 5 Microplastic degradation.

MPs' chronic effects can be passed along successive levels of the food chain, adversely harming species. The results of microplastics vary according to the organism species and the type and concentration of microplastics. (Sussarellu et al., 2016) have shown that polystyrene microplastics have a detrimental effect on oyster reproduction and eating by altering their food intake and energy allocation. When oysters were exposed to micro-sized polystyrene, their egg production, oocyte quality, and sperm motility decreased. Fertilization happens externally in the sea, where the eggs and sperm are released but due to the consumption of micro polystyrene, fertilization is harmed by decreased sperm speed and quantity (Sussarellu et al., 2016) (Sussarellu et al., 2016). 6- μ m micro polystyrene was discovered in the oyster's feces, with no accumulation in the gut, indicating a substantial polystyrene ejection. The output and growth of progeny of oysters exposed to microplastics decreased by 41% and 18%, respectively. The study provided data on the detrimental effects of micro-sized PS on the development and reproduction of oysters, with significant consequences for progeny. Energy allocation from reproduction to growth with a reduction in fertilization success results from polystyrene exposure investigations (Galloway et al., 2017). (Bessa et al., 2019) investigated the spread of microplastics in the Antarctic's aquatic ecosystem by examining the prevalence of microplastics in gentoo penguins. Although the water in Antarctic regions included microplastics, the concept of their absorption and entry food chain was examined. Seabirds, which have been recognized as biological monitors of environmental change, are also considered indications of ecological plastic pollution. Gentoo penguins' restricted movement outside their immediate surroundings makes them a standard indication for tracking plastic particles in Antarctic marine systems. The presence, identification, and characterization of microplastics in gentoo penguin scats. Penguin scats from two distinct islands were collected and found to include 58% microfibers, 26% pieces, and 16% films. Microplastics enter the gastrointestinal tract of penguins directly or indirectly through contaminated prey or polluted waters. Plastic waste accumulates in penguins' intestines, preventing them from consuming food and also causing them to absorb poisonous compounds from the water, impairing their growth and develop (Bessa et al., 2019). (Cole et al., 2015) demonstrated how MPs alter the feeding behavior, fertility, and function of zooplankton such as copepods (shown in Fig. 6).

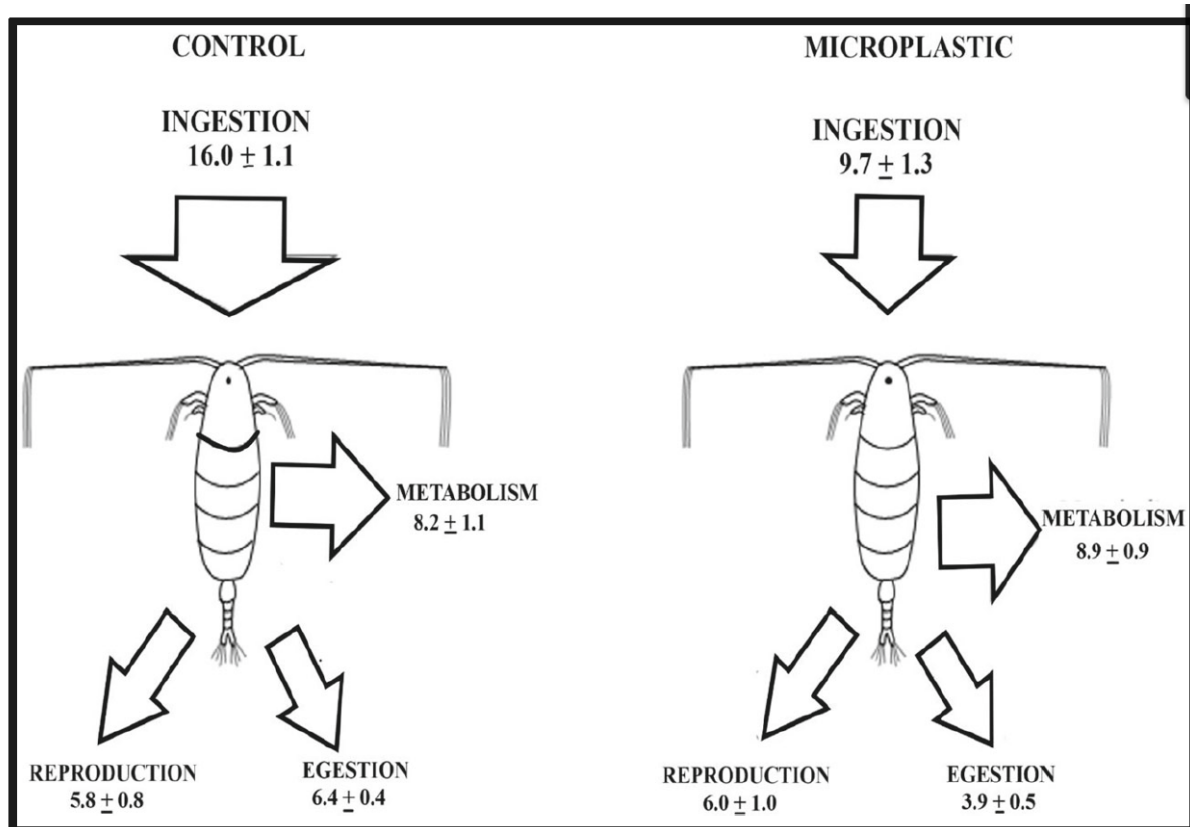


Figure 5: Effects of copepod in the controlled and microplastic medium.

The research was conducted on the *Calanus helgolandicus* copepod, which is found primarily in the Atlantic and is a key species that serves as prey for numerous fishes' larvae because of its enormous size, high lipid content, and luxury. Consumption of microplastics by copepods has a noticeable effect on their feeding, hatching, and overall health. Copepods subjected to polystyrene microbeads of 20 μ m resulted in a 40% drop in carbon biomass and an energy shortfall, indicating fast lipid consumption and impairing their growth. Copepods also die as a result of energy deficiency. Long-term exposure to microplastics results in tiny eggs with a low hatch rate (Cole et al., 2015). (Zocchi & Sommaruga, 2019) investigated the effect of microplastics on the toxicity of glyphosate, a herbicide. The tests were conducted on *Daphnia Magna*, which is very sensitive to microplastics and glyphosate. *Daphnia Magna* crustaceans provide food for various aquatic creatures and feed on microscopic particles in the water. Three different glyphosate formulations were tested, including glyphosate monoisopropylamine salt, glyphosate acid, roundup granules, and two other microplastics composed of beads and fibers. Without microplastics, the fatality rate for glyphosate-monoisopropylamine salt was high (23.3 percent), but when microplastics were added, a change in toxicity was found, with the highest mortality rate.

Additives	Properties	Effects	Reference
UV Stabilizers/absorbers	Prevents photodegradation	Mutagenic, toxic, bioaccumulated and show estrogenic activity	Hammer et al. 2012
Antioxidants	Delay oxidation, prevents aging	Estrogenic effects	Hermabessiere et al. 2017
Plasticizers	Renders the material pliable	Renal, reproductive, cardio, and neuro-toxicity	Rowdhwal and Chen 2018
Flame retardants	Diminish flammability	Endocrine disruptors	Fred-Ahmadu et al. 2020
Pigments	Colour	Duplication of food resulting gut blockage	Hammer et al. 2012
Surfactants	Modification of surface properties	Destroy mucus layer, damage gills	Rani et al. 2015

Figure 6: Chemicals found in microplastics effect of Addictives

UV Stabilizers/absorbers Defends against photodegradation Mutagenic, carcinogenic, bioaccumulative, and estrogenic. Antioxidants Prolongs the oxidation process and thereby prevents aging Estrogenic properties. Plasticizers Makes the substance more malleable Toxic effects on the kidneys, reproductive system, cardiovascular system, and nervous system Anti-flame agents Reduce flammability Disruptors of the endocrine system Pigments Colour Food duplication results in intestinal obstruction. Surfactants Changes in the surface qualities Destroy mucus layer, cause gill injury.

When mussels are exposed to microplastics, their grip is compromised due to a decrease in thread synthesis that aids in their adhesion. Mussels build reefs for protection and reproduction, and hence play a critical function in aquatic systems (Green, 2016). In (Berglund et al., 2019) investigated the effects of microplastics on mussels. An increase in the amount of microplastics was noticed as the mussel size increased. The large-sized mussel absorbs a lot of water while storing a lot of fibres. For plastic trash, the size-related filtering role is less obvious. The sum of fibres was more than the number of plastics, which could be explained by plastics' lower concentration or size. Increased microplastic concentrations result in faecal impaction or malnutrition in mussels. Additionally, plastics have a high capacity for attracting contaminants, which leads in the discharge of harmful compounds into the mussels (Berglund et al. 2019). For the first time, microplastics and fibres were discovered in the demersal shark species of the United Kingdom. Sharks obtain MP primarily through their diets, which are primarily crustaceans and molluscs, or through direct feeding. Due to the researchers' small particle size detection, there is a possibility of immediate excretion; however, the presence of chemicals bound to the fibres may have an effect on their reproductive cycle and immune systems (Parton et al., 2020).(Besseling et al., 2017)on how microplastic ingestion affects marine worms' vulnerability to pesticides (PCB). After 28 days of exposure to polyethylene, *Arenicola marina* displayed decreased feeding and growth, a high mortality rate, and bioaccumulation (Besseling et al., 2017).

Name	Polymer	Microplastic size	Number of test species	Experiment conditions	Impact	Reference
Nematode (Caenorhabditis elegans)	PS	1 μm	40	72 h, 0.45 NaOH, 2% HClO	Intestinal injury, oxidative stress	Yu et al. 2020
Zebra fish (Dania rerio)	PA PE PP PVC	70 μm	48	23 $\pm$ 1 $^{\circ}\text{C}$ 12 h photoperiod	Intestinal damage	Lei et al. 2018
Ascidian ciona intestinalis	PS	1 μm	30	18 $\pm$ 1 $^{\circ}\text{C}$ 8 days	Growth, food uptake	Messinetti et al. 2019
Sardinella gibbose (Fish)	PET PA	1 μm	25	Oscillation incubator at 65 $^{\circ}\text{C}$ at 80 rpm, 24 h, 1.2 g/mL NaCl	Feed habit, body weight	Hossain et al. 2019
Emys orbicularis (pond turtle)	PE	500–1000 mg kg^{-1}	36	1 month, 22 $\pm$ 2 $^{\circ}\text{C}$, photoperiod-14 light and 10 dark	Liver and kidney	Banaee et al. 2020
Crepidula onyx (Mollusca)	PS	2 μm	1000 larvae	22 $\pm$ 0.5 $^{\circ}\text{C}$, 12:12 light/dark cycle	Growth	Lo and Chan 2018

Figure 7: Study of microplastic impacts on various aquatic species

Figure 7 summarises the effects of microplastics on various species, including the polymer type, microplastic size, number of specimens used, and experimental conditions. Microplastics are identified at all stages of the food chain, impacting the gastrointestinal tracts and tissues, and their presence varies according to type and location. Due to their comparable size, organisms in the marine ecosystem mistook microplastics for food. Microplastics are ingested by all marine organisms, however the amount ingested varies by species. It is critical to monitor the excessive use of plastic additives and to adopt legislation and guidelines to regulate the origins of plastic litter, as MPs pose a threat to marine biota.

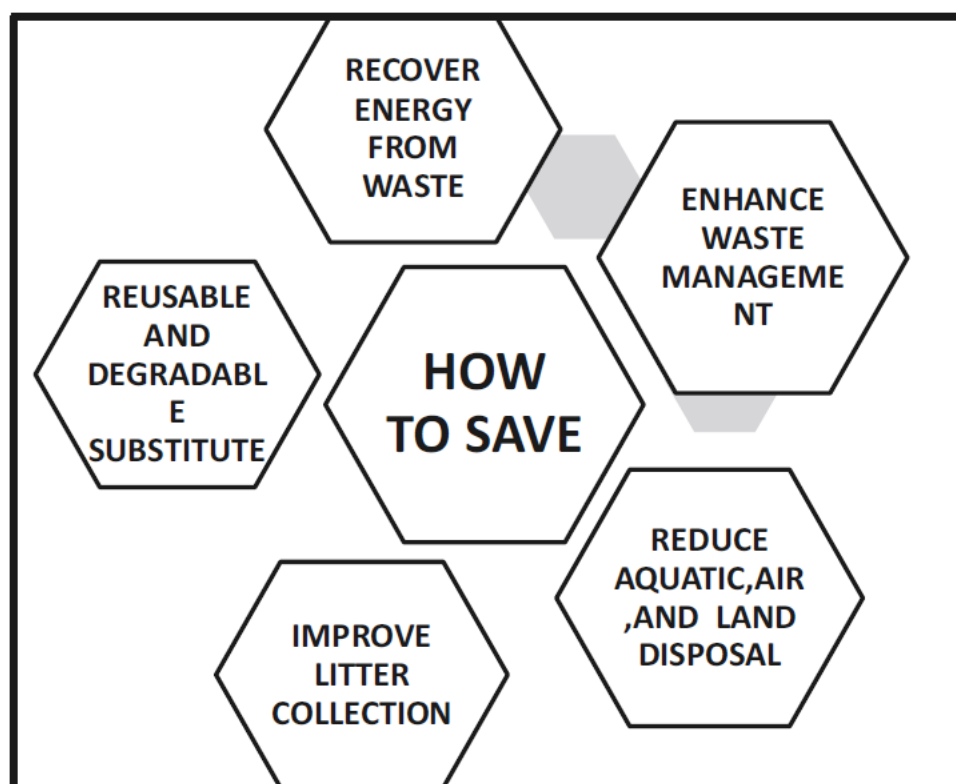


Figure 8: Strategies to reduce Microplastic pollution

Figure 8 illustrates the fundamental strategies that have been used to reduce pollution. Microplastic pollution research being conducted at the moment.

6. Recent studies on microplastic pollution

Microplastics have been discovered in the air we breathe (Enyoh et al., 2019), the food we eat (Eerkes-Medrano et al., 2019; D. Zhang et al., 2020), and the soil in which our crops thrive. Plastic particles have infiltrated even the world's most remote and pristine regions. According to a recent study (Ross et al., 2021), the Arctic is heavily contaminated by MP fibres, which could have come from clothing laundering. Due to increased textile manufacture and a lack of microfiber breakdown, microfiber aggregation will grow increasingly severe (Liu et al., 2021; Liu & Tokunaga, 2019). The most extensive investigation has revealed microplastics in 96 of 97 samples gathered from throughout the arctic seas. More than 92 percent of the MP was determined to be fibers, with 73 percent being polyester with the same size and colour as those found in clothing. Due to the reduced density of microfibers) can be transported over a greater distance via water and wind with high contamination of aquatic species than other types of microplastics. Due to the small size of microfibers, a significant amount can escape treatment plants and end up in marine environments.

Due to their non-corrosive qualities, glass-reinforced plastics are a new developing contaminant whose breakdown leads to the release of microplastic and fiberglass. When powdered GRP was applied to blue mussels, it revealed the presence of MP in the mantle cavity and inflammation in the gills. In contrast, in *Daphnia Magna*, it resulted f, increasing their average weight and allowing them to sink, thereby altering their swimming pattern (Ciocan et al., 2020). Numerous marine animals are at risk of ingesting or becoming entangled in a large amount of plastic trash poured into the water, and the mechanism by which species interact with plastic in the oceans remains a mystery. One possible explanation is that plastic resembles food, like when plastic bags are mistaken for jellyfish. According to a study on sea turtles, aquatic species are attracted to plastic garbage for its appearance and its smell. After several days in the waters, plastic garbage becomes covered in algae and other microorganisms and begins to smell like food. Sea turtles respond similarly to airborne olfactory receptors derived from bio-fouled plastics as they do to food odours which means that areas, where plastic wastes are concentrated, can act as olfactory traps, attracting the attention of other species and potentially causing harm. *Eurythenes plasticus*, a novel amphipod crustacean found in marine settings, was named after researchers discovered plastic garbage in its hindgut. The particle, which was around 649.648 metres in length and 83.74 percent identical to PET, a popular polymer used in bottles, food packaging, and textile textiles, was discovered in the species' guts. The shrimp-like creature, which measured about two inches in length and was discovered 20,000 feet underwater. Demonstrates that even creatures yet to be discovered are already polluted with plastics due to their life-long feeding on MPs, which may have acute and permanent health consequences. Although the ecotoxicological consequences of microplastics on deep-sea amphipods have not been explored, it is very feasible that the other undiscovered species dwelling in the Pacific Ocean's depths are also vulnerable to microplastic fibre consumption (Peng et al., 2020).

7. Conclusions and future implications

Microparticles, generally referred to as microplastics, are infesting the aquatic habitat on a global scale, where a diverse variety of creatures absorb these minute particles, which comprise a large portion of the population, producing chronic consequences. Plastics' significance as a carrier for poisonous compounds or as an abrasive that physically harms species in their habitat is significantly less obvious. Microplastic concentrations continue to rise in the ecosystem as a result of primary microplastic's continuous emergence and the decomposition of bigger plastics. Microplastic impacts exemplify a serious problem that affects nearly every marine system on the planet, regardless of how isolated it is from possible polluting sources. However, research on freshwater micropollutants is restricted in comparison to that on marine systems; however, it has become a source of great worry in recent years. MP found in freshwater is particularly significant for contamination accumulation due to its proximity to sources and reachability to more pollutants. As a result, animals living in freshwater settings may face increased exposure, particularly in the vicinity of industrial and densely populated areas, which may have higher concentrations of both hydrophobic toxins and microplastics.

Gammarus pulex, a freshwater crustacean, grew slower as the polystyrene content increased (Redondo-Hasselerharm et al., 2018)). To combat the damaging effects of microplastics, it is necessary to handle plastic trash in a cost-effective, high-quality, and environmentally friendly manner. While the primary source of plastic trash in water is terrestrial activities, it is advised that this issue be addressed. Garbage management strategies include reducing the number of sources, innovating products, reusing water, recycling and mulching, converting waste to energy, and preventing debris from entering the ocean at entry points. Microplastics may be used in place of metals such as aluminium and their combinations in the future. Because metals have adverse toxic effects on the human body when administered, the use of microplastics in medicines as an active and alternate ingredient may help treat diseases such as duodenal ulcer and gastritis. Microplastic removal and biodegradation from water and aquatic systems are presently limited in the laboratory, but could be marketed on a big scale in the future employing microorganisms such as fungi, protozoans, and bacterial spores. To prevent additional aquatic contamination, methods can be implemented prior to the conversion of microplastics to nanoplastics by segregating the high-density plastics that accumulate at the bottom from the low-density plastics that float at the surface of the water. This can be performed by incorporating coagulants; however, the adverse consequences of these substances on the human body are still unknown. Microplastics can be prevented from entering water bodies using sewage outlets fitted with ceramic filters. Additionally, given the involvement of the COVID-19 pandemic in microplastic contamination, there is a strong need for environmentally friendly solutions, which means that more research on biodegradable PPEs is needed to avert a potential microplastic pandemic.

8. Reference

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